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SUPERSEDING
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5 February 1979

MILITARY SPECIFICATION

COAGULANT, WATER TREATEMENT:

POLYELECTROLYTE, CATIONIC, LIQUID

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of liquid cationic, polyeletrolyte, water treatment coagulant.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood

MILITARY

MIL-D-43703 - Drum, Shipping and Storage, Molded Polyethylene.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 93	Flash Point by Pensky-Martens Closed Tester.
ASTM D 1177	Freezing Point of Aqueous Engine Antifreeze Solution.
ASTM D 2111	Specific Gravity of Halogenated Organic Solvents and Their Admixtures.
ASTM D 3951	Standard Practice for Commercial Packaging.
ASTM E 70	pH of Aqueous Solutions with Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Discription. The water treatment coagulant, hereinafter referred to as "coagulant", shall be a homopolymer, water soluble, viscous liquid, cationic polyelectrolyte.

3.2 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.2.1 Dimethyldiallyl ammonium cnloride (DMDAAC). The coagulant shall contain not less tnen 19.5 percent and not more than 20.5 percent homopolymer DMDAAC (see test method 30.1 in the appendix). The DMDAAC shall be in the molecular weight range of from 25,000 to 100,000.

3.3 Chemical and physical properties. The chemical and physical properties of the cogulant shall be as specified in table 1.

TABLE I. Chemical and physical properties of coagulant.

Chemical or physical property of coagulant	Minimum (min)	Maximum (max)
Total solids content, percent.	19.50	20.50
Viscosities at 25 °C (77 °F):		
Brookfield, in centipoises (cPs)	10.00	150.00
intrinsic, in deciliters per gram (dL/g)	0.14	0.26
Hydrogen-ion concentration (pH) value	3.20	3.80
Specific gravity, average at 25 °C (77 °F)	1.020	1.035
Flashpoint, °C (°F)	93 (200)	-
Freezing point, °C (°F)	-	-3 (27)

3.4 Unit containers. The coagulant shall be furnished in 5 gallon (gal) (19 liter [L]) unit containers conforming to MIL-D-43703, size 1, or 2.5 lb plastic bottles or 55 gallon drums. The unit containers shall be equipped with a flexible spout and an air vent. The flexible spout snall be closed with a screw cap and gasket. The air vent shall be closed with a buttress plug. After closure, the unit container shall not leak.

3.4.1 Unit container net weight. The net weight of coagulant in a unit container shall be not less than 40 pounds (lbs) (18.4 kilograms).

3.5 Identification marking. The unit containers shall be marked in accordance with MIL-STD-130. The nomenclature "Coagulant, Water Treatment: Cationic Polyelectrolyte" shall be included in the marking.

3.6 Workmanship. The coagulant shall be free from foreign matter, sediment or turbidity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in tne contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3)
b. Inspection of packaging (see 4.3)

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling shall be as specified in 4.3.1.1 through 4.3.1.3.

4.3.1.1 Inspection lot. For the purpose of quality conformance inspection, a lot shall be all of the coagulant produced by one manufacturer at one plant, from the same batch of raw materials and by the same manufacturing process, which is offered for inspection at one time. The lot size shall be determined as the quantity of filled unit containers offered for inspection.

4.3.1.2 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.1.3 Sampling for tests. Two filled unit containers shall be selected from each inspection lot for test.

4.3.2 Examination. Samples selected as specified in 4.3.1.2 shall be examined as specified in 4.4.1. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.3.3 Tests. The sample selected in accordance with 4.3.1.3 shall be tested as specified in table II, in the order shown. Failure of any test shall be cause for rejection of the lot.

TABLE II. Test Schedule.

Test No.	Test	Test paragraph	Requirement paragraph
1.	Percent DMDAAC in coagulant.	4.4.2.1	3.2.1
2.	Total solids content.	4.4.2.2	3.3, table I
3.	Brookfield visosity.	4.4.2.3	3.3, table I
4.	Intrinic viscosity.	4.4.2.4	3.3, table I
5.	pH level.	4.4.2.5	3.3, table I
6.	Specific gravity.	4.4.2.6	3.3, table I
7.	Flashpoint.	4.4.2.7	3.3, table I
8.	Freezing point.	4.4.2.8	3.3, table I

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4.4 Inspection procedure.

4.4.1 Examination. The coagulant shall be examined for the following defects:

Defect No.	Defect	Requirement paragrapn
Major		
101.	DMDAAC in coagulant not of specified molecular weight.	3.2.1
102.	Unit containers not as specified, not closed as specified, or leaking.	3.4
103.	Net weight of coagulant in unit containers not as specified.	3.4.1
Minor		
201.	Identification marking on unit containers not as specified.	3.5
202.	Workmanship not as specified.	3.6

4.4.2 Tests.

4.4.2.1 Percent DMDAAC test. The percentage of DMDAAC in the coagulant shall be determined in accordance with test procedure 30.1 of the appendix to this specification. A DMDAAC percent that is less than 19.5 or a DMDAAC percent that is greater than 20.5 shall constitute failure of this test.

4.4.2.2 Total solids content test. Weigh an evaporating dish to the nearest 0.0001 gram (g). Record the weight of the dish as W_1. Add 5 to 10 g of coagulant to the dish and immediately weigh the dish and contents. Record this weight as W_3. Place the dish and contents in a preheated, constant temperature oven set at 105 °C (22 °F) Allow the coagulant to dry in the oven overnight, but not less than 16 hours. After the drying period, remove the dish and contents from the oven and place in a dessicator to cool. When cool, weigh the dish and contents to the nearest 0.0001 g. Record the weight as W_3. Calculate the total percent of solids in the coagulant using the following equation:

$$\text{Percent total solids} = \frac{W_3 - W_1}{W_2 - W_1} \times 100$$

A total solids content of less than 19.5 percent or more than 20.5 percent shall constitute failure of this test.

4.4.2.3 Brookfield viscosity test. Pour approximately 300 milliliters (mL) of coagulant into a 400 mL beaker. Adjust the temperature of the coagulant in the beaker to exactly 25 °C (77 °F), and maintain the temperature throughout the test period. Attach a number 1 bob to a Brookfield viscometer, model LVF. Immerse the bob in the coagulant to the notch on the bob stem. Connect the

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viscometer to a power source and turn the viscometer on. Set the viscometer to operate at 30 revolutions per minute (rpm). After allowing the viscometer to operate for several seconds in order to stabilize rpm, depress the viscometer clutch and read the torque drag on the hundredth scale. Convert the reading obtained to cPs by means of the conversion chart furnished with the viscometer. A Brookfield viscosity of less the 10.0 cPs or more than 150.0 cPs shall constitute failure of this test.

4.4.2.4 Intrinsic visosity test. The intrinsic viscosity of the coagulant shall be determined in accordance with test procedure to this specificaitoin. An intrinsic viscosity of less than 0.14 dL/g, or more than 0.26 dL/g, shall constitute failure of this test.

4.4.2.5 pH level test. The pH level of the coagulant shall be determined by the calomel and glass electrode (electrometric) method of except that the assembly shall be standardized with only one buffer solution within 1 to 2 units of the normal coagulant pH. The actual pH result shall be recorded to the nearest tenth of a unit. A pH level of less than 3.20 or more than 3.80 shall constitute failure of this test.

4.4.2.6 Specific gravity. The specific gravity of three specimens from each sample shall be determined in accordance with, ASTM D 2111 method B. An average specific gravity of less than 1.020 or more than 1.035 at 25 °C (77 °F) shall constitute failure of this test.

4.4.2.7 Flashpoint test. The flashpoint of the coagulant shall be determined in accordance with ASTM D 93. A recorded flashpoint that is less than 93 °C (200 °F) shall constitute failure of this test.

4.4.2.8 Freezing point test. The flashpoint of the coagulant shall be determined in accordance with ASTM D 1177. A freezing point of the coagulant that is more than -3 °C (27 °F) shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformace inspection of packaging.

4.5 1.1 Unit of product. For the purpose of inspection, a completed pack, prepared for shipment, shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 1 0 percent defective.

No.	Defect	A	B	Comm
104.	Shipping container not close-fitting as specified.	5.2.1	5.2.2	
105.	Shipping container not as specified.	5.2.1	5.2.2	5.2.3

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106.	Strapping not as specified.	5.2.1	5.2.2	
107.	More than 8 unit containers placed in shipping container.	5.2.1	5.2.2	
108.	Marking not as specified.	5.3.1	5.3.1	5.3.2

5. PACKAGING

5.1 Preservation. The coagulant shall be placed in the unit container specified in 3.4. The container shall be considered to satisfy the needs for level A preservation.

5.2 Packing. Packing shall be level A, level B or commercial as specified (see 6.2).

5.2.1 Level A. The coagulant, in the specified unit containers (see 3.4), shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Box closure and strapping shall be as specified therein except that the strapping shall be flat and the finish B. Not more than 8 unit containers shall be packed in a shipping container.

5.2.2 Level B. The coagulant shall be packed as specified for level A in 5.2.1 except that the strapping may be finish A.

5.2.3 Commercial. The coagulant, in the specified unit containers (see 3.4), shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Levels A and B. In addition to the markings required in 3.5 and any special or identification markings required by the contract or purchase order, all containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. Additionally, each shipping container shall be marked with the cube and gross weight.

6. NOTES

6.1 Intended use. The coagulant is intended for use with military reverse-osmosis water treatment equipment.

6.2 Ordering data. Acquisition documents should specifiy the following:

a. Title, number and date of this specification.
b. Quantity of unit containers of coagulant required.
c. Degree of packing required (see 5.2).
d. Any special marking (see 5.3.1).

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6.3 Subject term (Key words) listing.

Coagulant
Polyelectrolyte

6.4 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activity:
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APPENDIX

10. SCOPE

10.1 Scope. This appendix details the test procedures required to ascertain the percent of DMDAAC in coagulant by ionic chloride concentration determination, and the intrinsic viscosity of the coagulant.

20. APPLICABLE DOCUMENTS

20.1 Publications. The following documents form a part of this appendix to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 455	- Test for Kinematic Viscosity of Transparent and Opaque Liquids.
ASTM D 446	- Specification and Operating Instructions for Glass Capillary Kinematic Viscometers.
ASTM E 1	- Specification for ASTM Thermometers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

30.1 Percent DMDAAC test.

30.1.1 Test equipment. Test equipment to determine the percent of DMDAAC in the coagulant shall be as follows:

1. Balance, anlytical, sensitive to 0.0001 g.
2. Beakers, glass, 250 mL.
3. Beaker, polypropylene, 150 mL.
4. Buret, 25 mL, calibrated to 0.1 mL.
5. Flask, volumetric, polyproylene, 1 L.
6. Flask, volumetric, polypropylene, 2 L.
7. Flask, volumetric, glass, 1 L.
8. pH meter, pH/millivolt (mv) type, - 1000 mv to + 1000 mv range, with appropriate electrodes.
9. Paper, semi-logarithmic, 1 cycle.
10. Pipet, Eppendorf type, 500 microliters (l).
11. Pipet, Mohr type, glass, 1 mL.
12. Pipet, Mohr type, polypropylene, 1 mL.
13. Pipet, Mohr type, polypropylene, 5 mL.
14. Pipet, volumetric, glass, 100 mL.
15. Pipet, volumetric, polypropylene, 100 mL.
16. Pipet, volumetric, polypropylene, 200 mL.
17. Spectrophotometer, Bausch & Lomb model Spectronic 20 or equal, with 1/2-inch (12.5 millimeter [mm]) cells.

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APPENDIX

30.1.2 Reagents. Reagents to determine the percent DMDAAC in the coagulant shall be as follows (see Note A):

1. Sodium buffer solution: Calgon Corporation catalog No. CR 2066 or equal.
2. Dilute nitric acid solution: prepare by adding 50 mL of concentrated nitric acid (specific gravity [Sp. Gr.] 1.42) to about 700 mL of distilled water in a 1 L volumetric flask. Dilute to the mark with distilled water and mix thoroughly.
3. Diphenylcarbazone solution: prepare by dissolving 0.3 g of crystalline diphenycarbazone in 100 mL of absolute methyl alcohol (see Note B)..
4. Standard mercuric nitrate solution: prepare as follows:

   a. Dissolve approximately 2.8 g of reagent grade mercuric nitrate monohydrate in exactly 0.5 mL of concentrated nitric acid (Sp. Gr. 1.42) and about 50 mL of distilled water in a 1 L volumetric flask. Dilute the solution to the mark with distilled water and mix thoroughly. Allow the solution to stand for not less than 24 hours before it is standardized.
   b. Prepare a stock chloride soluion by dissolving 1.6 g, weighed to the nearest 0.0001 g, of pure sodium chloride that has been dried for not less than hour at 105 °C (221 °F), in 800 ml of distilled water in a 1 L polypropylene volumetric flask. Dilute to the mark with distilled water and mix thoroughly. Record the weight of the sodium chloride as W. Correct the weight for any impurity of the sodium chloride if the assay was less than 99.8 percent.
   c. Prepare a standard chloride solution by diluting 100 ml of the stock chloride solution into 1 L of distilled water. Determine normality of the standard chloride solution by pipetting 100 ml of the standard chloride solution and 10 drops of diphenylcarbazone solution (see Note C).
   d. To standardize the mercuric nitrate solution, titrate a 100 mL sample to the endpoint with standard chloride solution from a 25 mL buret. The color will change from light yellow to light purple to deep blue-violet. Determine endpoint of titration by measuring the color of a portion of the sample at set intervals in a 1/2-inch (12.5 mm) cell placed in the spectrophotometer which has been set to a wavelength of 520 nanometers (nm). Repeat the standardization until 3 values that differ by not more than 0.05 mL are obtained. Add the 3 values obtained and subtract the test blank.
   e. The value of the test blank is usually 0.1 mL to 0.2 mL. The exact value of the test blank should be determined by titrating 100 mL of chloride-free water, to which 1 drop of dilute nitric acid solution and 10 drops of dilute nitric acid solution and 10 drops of diphenylcarbazone solution have been added.

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f. Normality of the standard mercuric nitrate solution should be 0.0175. Calculate normality by the following equation:

$$\text{Normality of standard mercuric nitrate solution} = \frac{(W)}{(V_s - V_1)\ (0.585)}$$

Where:

W = weight of NaCl in grams
V_s = average mL of titrant required for chloride solution
V_1 = average mL of titrant required for test blank

$$0.585 = \frac{58.5 \text{ gm/mole NaCl}}{100 \text{ mL NaCl titrated}}$$

5. Standard sodium solutions: prepare 100 parts per million (ppm) stock solution and 10 ppm, 2 ppm and 1 ppm standard sodium solutions as follows (see Note D):

a. 100 ppm stock sodium solution: prepare by dissolving exactly 0.510 g of pure sodium chloride in 1 L of distilled water in a 2 L polypropylene volumetric flask. Dilute to the mark with distilled water and mix thoroughly.
b. 10 ppm sodium solution: prepare by diluting exactly 100.0 mL of the 100 ppm stock sodium solution in 1 L of distilled water in a 1 L polyproylene volumetric flask. Add 2.5 mL of sodium buffer solution by Mohr-type pipet and mix thoroughly.
c. 2 ppm sodium solution: prepare by diluting exactly 200.0 mL of the 10 ppm sodium solution in 1 L of distilled water in a 1 L polypropylene volumetric flask. Add 2.5 mL of sodium buffer solution by Mohr-type pipet and mix thoroughly.
d. 1 ppm sodium soluton: prepare by diluting exactly 100.0 mL of the 10 ppm sodium solution in 1 L of distilled water in a 1 L polyproylene volumetric flask. Add 2.5 mL of sodium buffer solution by Mohr-type pipet and mix thoroughly.

NOTES:

A. To obtain results accurate to within ±1 percent, good laboratory technique is required. Pipets and burets should be cleaned periodically with cleaning solution to eliminate droplet holdup on the surface of the glass. Do not use a pipet or buret with a broken tip. After leveling a solution to the mark on a pipet or buret, make sure the drop on the tip is removed. After draining a pipet or buret, touch the tip to the interior of the beaker or flask, so that similar drainage to less than one drop is obtained.
B. The diphenylcarbazone solution will remain stable for a period of from 4 to 6 months if stored in an amber bottle.

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APPENDIX

C. The standard mercuric nitrate solution should be stored in an amber glass bottle away from light. Under such conditions, the solution has been found to change as little as 1 percent a month. For accurate results, restandardization at least once a week is stongly suggested. More frequent restandardization may be found necessary if the solution is not stored inside a dark cupboard immediatley after use. Do not store the solution in a polyethylene bottle.

D. Equipment used to prepare and store both stock and standard sodium solutions shall be all polypropylene. Containers and equipment shall be prerinsed before use by a 1 percent hydrogen chloride solution followed by water until the equipment is neutral.

WARNING

CONCENTRATED NITRIC ACID IS AN EYE AND SKIN IRRITANT. PURE MERCURIC SALTS ARE EXTREMELY TOXIC. GLOVES AND SAFETY GLASSES SHOULD BE WORN WHEN WORKING WITH THESE COMPOUNDS AND ALL WORK SHOULD BE PERFORMED UNDER A HOOD. MERCURIC SALTS SHOULD BE DISPOSED OF IN ACCORDANCE WITH REGULATIONS OF THE ENVIRONMETAL PROTECTION AGENCY AND OTHER LOCAL GOVERNMETAL JURISDICTIONS.

30 1.3 Sodium content determination. Determine the sodium content of the coagulant as follows:

1. Stabilize a sodium electrode and a sodium reference electrode in the 1 ppm standard sodium solution for 3 hours.
2. Fill the 150 mL polypropylene beaker with 1 ppm standard sodium solution. Equip the pH meter with the electrodes from step 1 and switch to the mv range. Lower the electrodes into the beaker and swirl to wet the electrodes.
3. After 1 minute, note the mv reading on the meter. Do not mix the standard solution during the 1 minute period.
4. Repeat step 3 until two consecutive readings agree to within ±0.5 mv. Record the mv reading.
5. Repeat steps 1 through 4 with 2 ppm and 10 ppm standard sodium solutions.
6. Prepare a calibration curve by plotting the recorded mv readings versus the ppm of sodium for the 1 ppm, 2 ppm, and 10 ppm standard sodium solutions on semi-logarithmic paper as illustrated in figure 1 to this appendix. Draw a straight line between the 1 ppm and 10 ppm points. If the 2 ppm point is more than 1 mv from the line, the electrodes were not properly stabilized and the procedure must be repeated. If the 2 ppm point is more than 1 mv off the line, the electrodes are fouled and new electrodes must be obtained and the procedure repeated until the 2 ppm point is within 1 mv of the 1 ppm to 10 ppm line.

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7. Pipet 100 mL of coagulant into a 150 mL polypropylene beaker using a 1 mL, Mohr type, polypropylene pipet. Add 0.25 mL of sodium buffer solution to the coagulant and mix. Repeat steps 2, 3, and 4 using the buffered coagulant in place of the standard sodium solution.
8. Determine the ppm of sodium in the buffered coagulant by referring to the mv reading of the meter to the calibration curve. If the buffered coagulant contains more than 2.5 ppm sodium, dilute it so that the sample will contain between 1.0 ppm and 2.5 ppm of sodium and repeat step 7 and this step.
9. Record the ppm sodium from the graph, corrected for any dilutions made, as C.

NOTE:

E. The 2 ppm point check described in step 6 shall be performed after every other coagulant sample sodium determination to assure that the electrodes have not fouled or drifted.

30.1.4 Percent DMDAAC determination. Ascertain the percent DMDAAC in the coagulant by ionic chloride determination as follow:

1. Weigh a calculated amount of coagulant to be tested according to the following equation:

$$\text{Approximate coagulant test weight} = \frac{40}{\text{expected percent}}$$

and record the actual weight of coagulant as A.

2. Transfer this quantity to a 1 L glass volumetric flask and dilute to the mark with distilled water.
3. Measure two 100 mL portions of the diluted coagulant into separate, clean 250, mL beakers with the 100 mL pipet.
4. Determine the portions of the diluted coagulant using the pH meter set on the pH reading and using a glass pH electrode. Add nitric acid solution one drop at a time until the meter indicates a pH value of from 2.8 to 3.2 has been attained. Count and record the number of drops that were required to obtain the specified pH value. Now, discard this solution of coagulant.
5. Add the number of drops of nitric acid solution recorded in step 4 into the other portion of dilute coagulant. Stir to mix thoroughly.
6. Add 10 drops of diphenylcarbazone solution to the mixture.
7. Titrate the preparation from step 6 with standard mercuric nitrate solution from a 25 mL buret. The color of the specimen will change from a light yellow to a light purple and then, to deep blue violet. Determine the end point of titration by measuring the color of a portion of the specimen at set intervals in a 1/2-inch (12.5 mm) cell placed in the spectrophotometer which has been set to a wave-

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length of 520 mm. Record the mL of titrate required to attain an absorbance of 0.28 to 0.32 (transmittance of 52 percent of 48 percent).

8. Calculate the percent DMDAAC as monomer or polymer from the following equation:

$$\text{Percent DMDAAC} = \frac{(V_1 - V_2)\ (N)\ (0.1615)\ (1000)}{A} - \frac{(C)\ (161.5)}{(230)\ (A)}$$

Where:

A = Recorded weight of specimen (step 1)

V_1 = mL of standard mercuric nitrate solution required to titrate the specimen.

V_2 = mL of standard mercuric nitrate solution required to titrate the blank (see Note F)

N = normality of standard mercuric nitrate solution

C = ppm sodium in coagulant determined in 30.1.3

NOTE:

F. Usually the value of the blank is 0.1 to 0.2 mL. The exact value should be determined by titrating with the standard mercuric solution into 100 mL of chloride-free water which has 1 drop of dilute nitric acid solution and 10 drops of diphenyl carbazone solution added.

30.2 Intrinsic viscosity test.

30.2.1 Test equipment. The test equipment required to determine the intrinsic viscosity of the coagulant shall be as follows:

1. Viscometer, Cannon-Fenske type, calibrated, size 50 in accordance with ASTM D 446.
2. Bath, viscometer, kinnematic precision type, minimum range of 20 °C (68 °F) to 121 °C (249.8 °F) and uniform to ±0.01 °C (0.018 °F) in accordance with ASTM D 445.
3. Thermometer, partial immersion type, number 36C in accordance with ASTM E 1.
4. Flask, volumetric, 100 mL.
5. Flask, filtering, side tube type, 260 mL.
6. Funnel, Buchner type, medium fritted, 30 mL.
7. Filter vac, 7/8-inch (22.23 mm) bottom opening.
8. Stopwatch, graduated to 0.01 second (sec).
9. Bulb, rubber, safety pipettor.

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30.2.2 Reagent. Reagent to determine the intrinsic viscosity of the coagulant shall be as follows:

1. Sodium nitrate solvent: prepare the solvent by adding 85 g of sodium nitrate to a 1 L volumetric flask containing about 800 mL of distilled water. Swirl to dissolve. Dilute to the mark with distilled water and mix thoroughly.

30.2.3 Intrinsic viscosity determination. Determine the intrinsic viscosity of the coagulant as follows:

1. Set the water bath temperature to 25 °C (77 °F) and allow sufficient time for equilibration.
2. Calculate the approximate weight of coagulant to be tested using the following equation:

$$C_A,\ g/dl = \frac{1}{2\,[\eta]_E}$$

Where: C_A = approximate weight of polymer, in g, to be dissolved in 100 mL of sodium nitrate solvent.

$[\eta]_E$ = expected intrinsic viscosity.

3. Weigh the sample specimen of coagulant to the nearest mg into a 100 mL volumetric flask. Record this weight as W. Correct W for moisture, solids, residual monomer, etc., to find actual weight of polymer added to the flask. Record actual polymer weight as C.
4. Add about 60 mL of distilled water to the flask and mix thoroughly until the coagulant is completely dissolved.
5. Add 8.5 g of sodium nitrate to the flask. Dilute to the mark with distilled water and mix thoroughly.
6. Filter about 100 mL of the sodium nitrate solvent through a medium porosity filter in the Buchner funnel into the clean, dry filtering flask.
7. Pour 10-20 mL of this filtered sodium nitrate solvent into the clean, dry viscometer tube. Place the tube in the water bath and allow time for equilibration at 25 °C (77 °F).
8. After thermal equilibration is attained, use the rubber bulb to draw solvent up and down the capillary three times. Draw solvent up the capillary a fourth time and allow it to drain back down by gravity. Measure the time required for the solvent meniscus to pass between the two marks on the viscometer tube and record.

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9. Discard the solvent and repeat steps 7 and 8 with fresh filtered solvent, again timing the descent of the meniscus between the marks. Continue to repeat this until two consecutive times agree to within ±0.2 second. Record the average time, in seconds, as t_o.
10. Discard the solvent. Do not clean or dry the viscometer tube.
11. Repeat steps 7 and 8 using the prepared sample from step 5. Repeat the process as indicated in step 9, except recording the average time, in seconds, as t (see Note G).
12. Calculate the intrinsic viscosity of the coagulant using the following equation:

$$[\eta]^{25\ ^\circ C}_{1N\ NaNO_3},\ dl/g = \left[2\left(\frac{t-t_o}{t_o C} - \frac{2.303 \log \left(\frac{t}{t_o} \right)}{C} \right) \Big/ C \right]^{1/2}$$

NOTE:

G. For repeated tests, the viscometer tube may be washed once with filtered sodium nitrate solution, and the next prepared coagulant sample tested.

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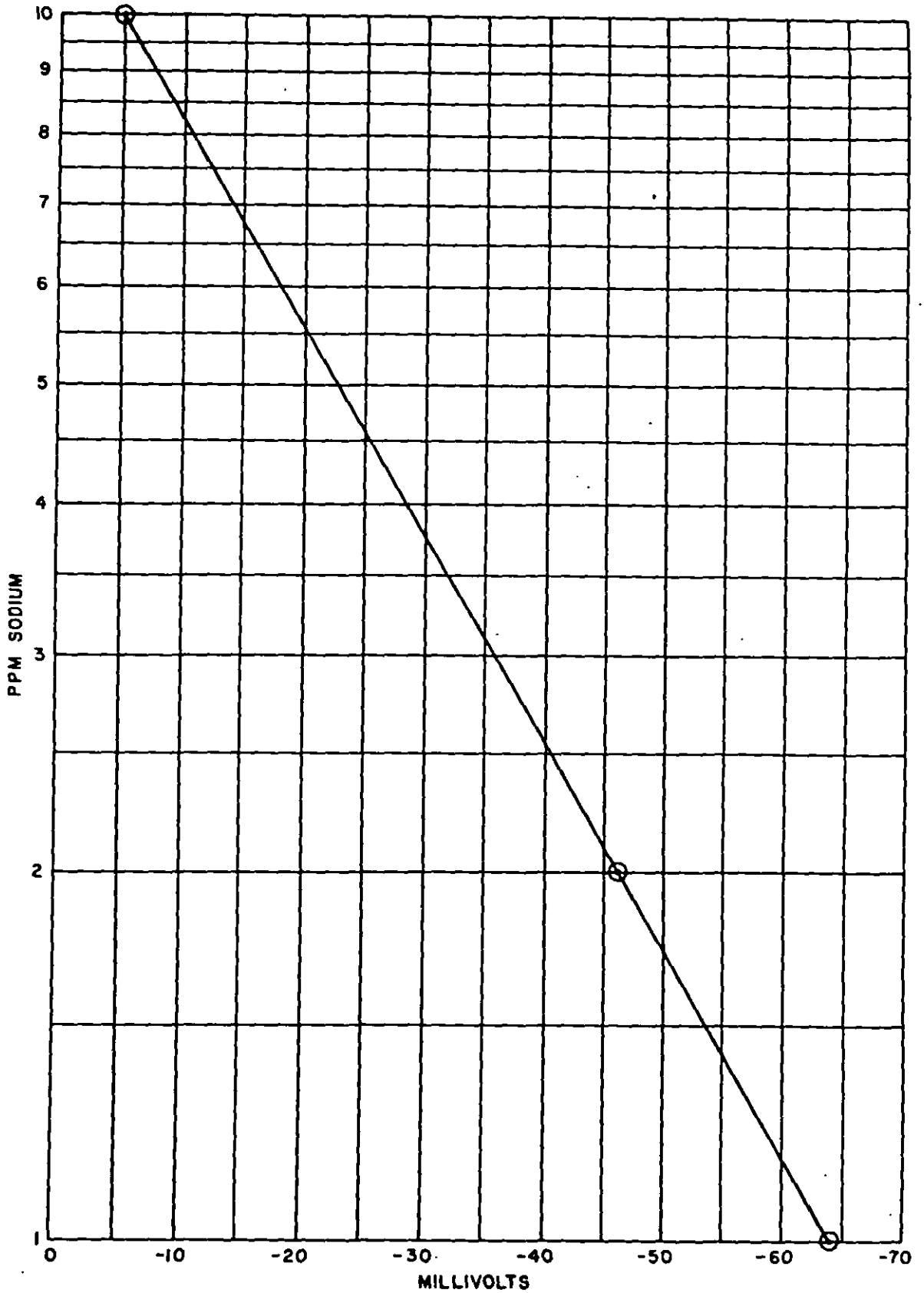


FIGURE 1. Calibration curve.

X-4319

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